

Alexus Huang

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Education

University of California, Santa Cruz

Sep. 2025 – Expected Jun. 2029

B.S. in Computer Science, GPA: 3.78

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Linear Algebra, Object-Oriented Programming, Computer Systems & Assembly, Programming Abstractions

Technical Skills

Languages: Python, JavaScript, SQL, HTML/CSS, Java, RISC-V Assembly

Frameworks & Libraries: Flask, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Chart.js

AI & APIs: Claude API, Model Context Protocol (MCP), REST APIs

Tools & Platforms: Git, GitHub, SQLite, Jupyter Notebook, PythonAnywhere

Projects

SlugStudySpots | Full-Stack Web Application

Live: slugstudy.pythonanywhere.com

Python, Flask, SQLite, Flask-Login, JavaScript, HTML/CSS

github.com/alexus-Huang/SlugStudySpots

- Built and deployed a full-stack Flask/SQLite web app for UCSC students to share study spots; 300+ total page visits.
- Implemented secure authentication (signup, login, sessions) with Flask-Login and Werkzeug password hashing, plus regex validation to block malformed and duplicate accounts.
- Prevented SQL injection with parameterized queries and built an interactive campus map for browsing spots.

Predictive CPU Load Monitor | ML Forecasting & Live Dashboard

Python, Flask, scikit-learn, psutil, Pandas, Chart.js, JavaScript

github.com/alexus-Huang/predictive-thermal-monitor

- Built an ML pipeline that logs live CPU/RAM telemetry with psutil and trains a scikit-learn linear regression to forecast CPU load 10 seconds ahead (held-out MAE 7.05 vs. ~11-point std. dev. baseline).
- Engineered rolling-average and short-term trend features (MAE 14.52 → 13.56), then compared Linear Regression against Random Forest and ran a controlled disk-I/O feature test, keeping the simpler model when neither helped.
- Served live predictions through a Flask REST API that buffers recent readings server-side to reproduce training features, feeding a Chart.js dashboard that polls every 3 seconds and flags NORMAL/WARNING status.

Gym Workout Analyzer | Data Cleaning & Visualization

Python, Pandas, Matplotlib, Seaborn, Jupyter

github.com/alexus-Huang/gym-workout-analyzer

- Built a Pandas data-cleaning pipeline that turned a raw workout log with inconsistent dates, mixed kg/lbs units, and duplicate or incomplete rows into an analysis-ready dataset.
- Standardized exercise names (e.g., "BP" → "Bench Press") and unified kg/lbs so lifts could be compared across weeks.
- Visualized 10 weeks of strength-training data with Seaborn/Matplotlib to surface lift-by-lift progression trends, identifying Deadlift and Squat as the fastest-progressing lifts and Overhead Press as a target for focused training.

Experience

Swim Instructor

Oct. 2024 – Apr. 2025

American Swim Academy

Fremont, CA

- Coached groups of up to 4 youth swimmers per class using structured, skill-level-specific lesson plans.
- Communicated progress and safety guidance to parents and students in a high-responsibility setting.

Secretary

Sep. 2023 – May 2025

Washington High School Key Club

Fremont, CA

- Recorded meeting minutes and maintained membership records for a 100+ member club.
- Coordinated logistics and communications for school and community service events.

Certifications

Anthropic: Building with the Claude API • Introduction to Model Context Protocol (MCP) • Claude Code 101

Google: AI Essentials Certificate • AI Professional Certificate • **UC Berkeley:** ROAR Certificate of Completion